

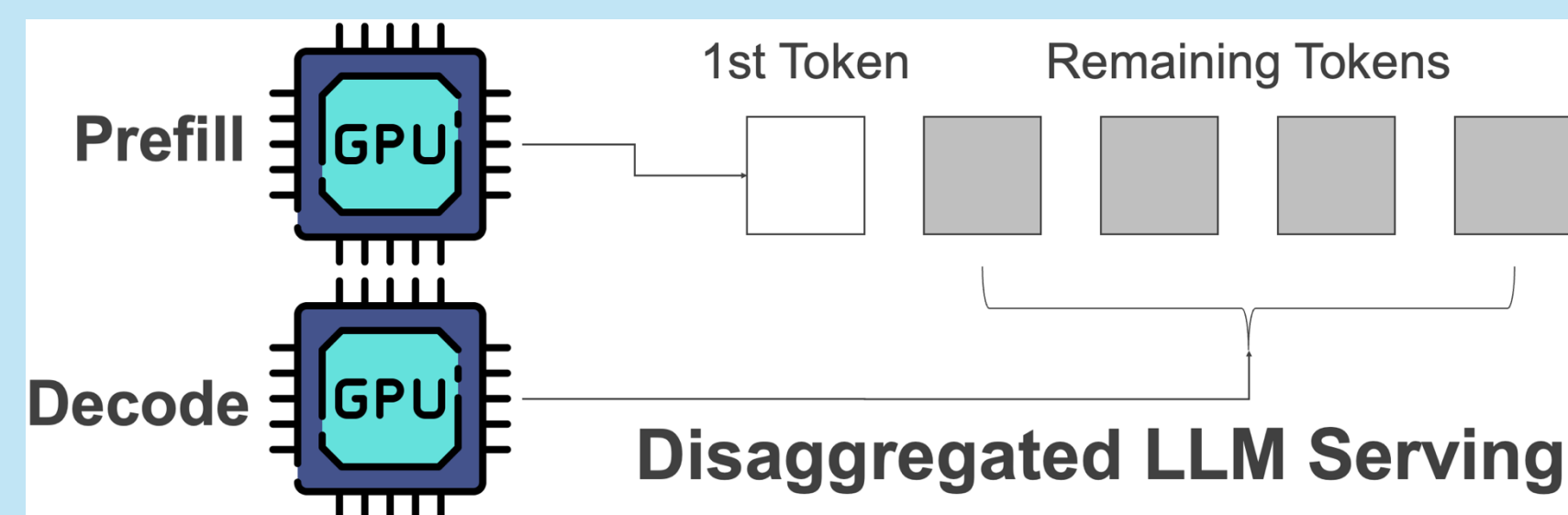
#77 Revisiting Disaggregated LLM Serving for Performance and Energy Implications

Jiaxi Li¹, Yue Zhu², Bo Chen¹, Eun Kyung Lee², Klara Nahrstedt¹, Presenter: Kiran Hombal¹

¹University of Illinois Urbana-Champaign, ²IBM Research

Contact: Jiaxi Li | Email: jiaxili3@illinois.edu

BACKGROUND AND BENCHMARK MOTIVATION



Prefill stage processes input tokens, generates KV cache, and generates the first output token
Decode stage autoregressively generates output tokens based on KV cache

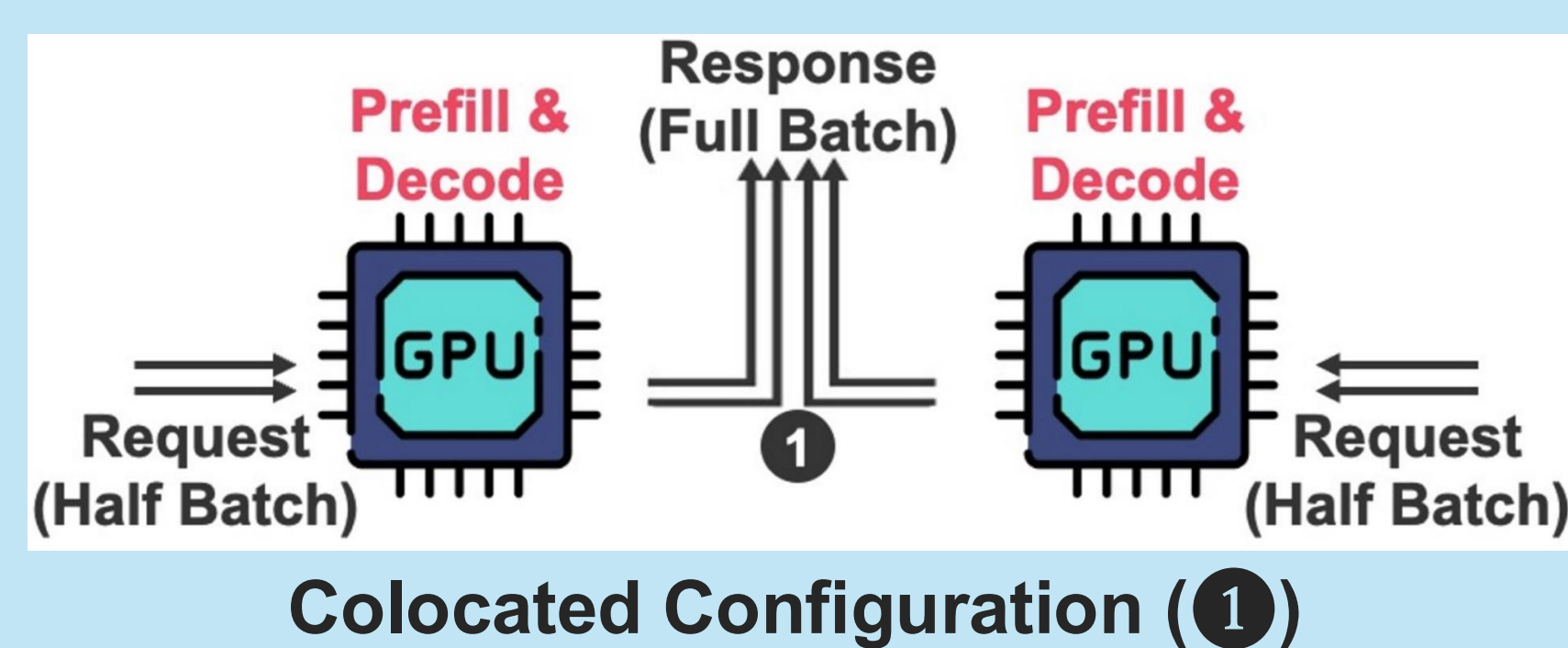
Background

- Colocated LLM serving places both prefill and decode stages on the same GPU, which leads to interference between the two stages [1]
- Disaggregated LLM serving assigns prefill and decode stages to separate dedicated GPUs, which requires KV cache transfer [2]

Motivation for Benchmarking

- KV Transfer Paths**
 - Different KV transfer paths (e.g., NVLink, PCIe, and disk offloading) exhibit trade-offs between transmission bandwidth and cost
 - Serving performance (e.g., latency and throughput) and energy consumption differences across KV transfer paths remain unclear [3]
- Disaggregated Frequency Scaling**
 - Real-world LLM requests exhibit imbalanced Service Level Objectives (SLOs) between the prefill and decode stages [4]
 - Dynamic voltage and frequency scaling (DVFS) enables trade-offs between power consumption and serving performance
 - Disaggregation allows stage-wise independent frequency scaling, further improving energy efficiency under per-stage SLO constraints

BENCHMARK METHODOLOGY



KV Transfer Path Configurations

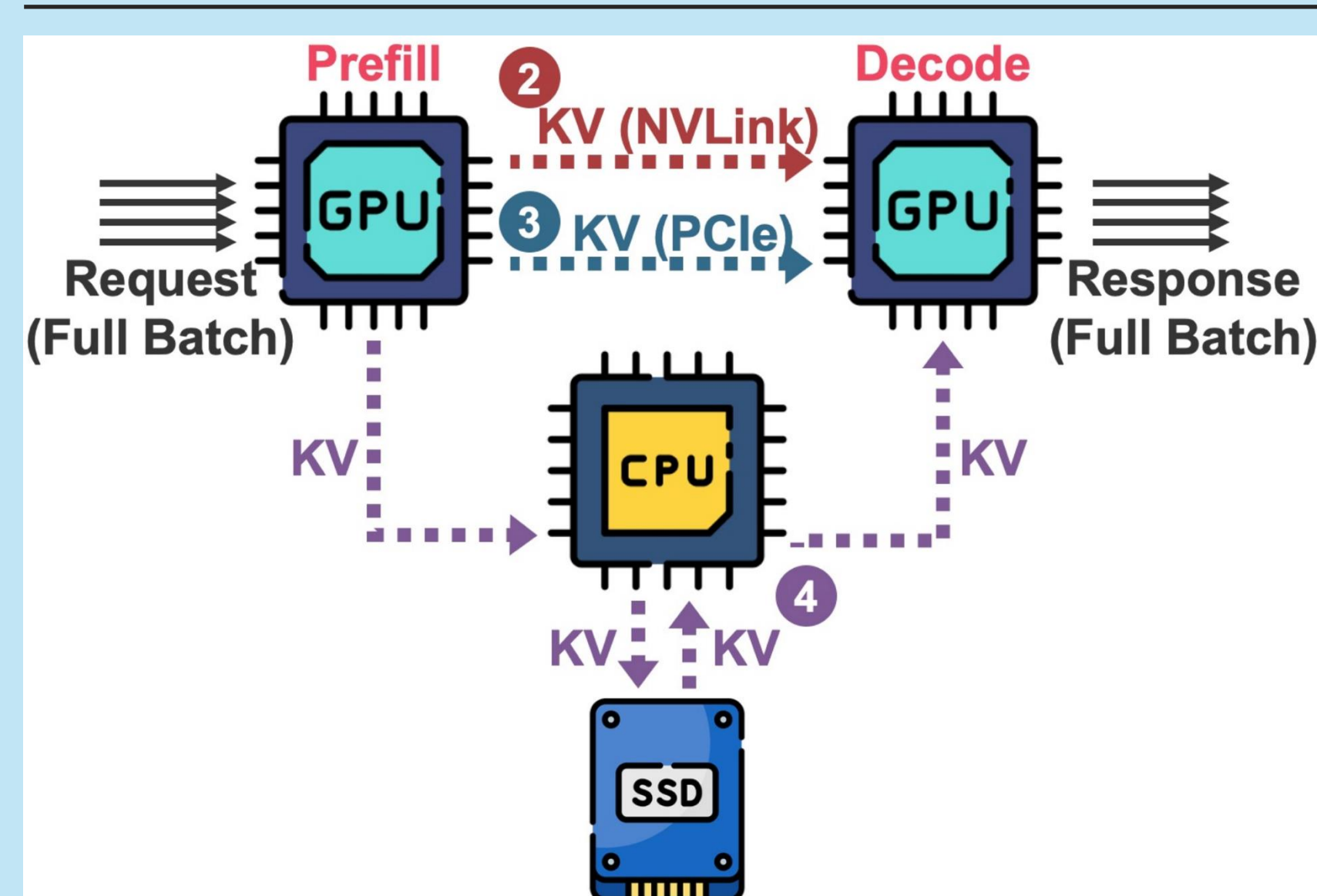
- co-2gpus**: Colocated serving on two GPUs each processing half batch
- dis-nv**: Disaggregated serving with KV transfer over 12 NVLink links
- dis-pcie**: Disaggregated serving with KV transfer over a PCIe x16 bridge
- dis-disk**: Disaggregated serving with KV transfer via disk offloading

Hardware

- GPU: 2 x NVIDIA A100 40 GB
- CPU: Intel Xeon E5-2697 v4
- DRAM: Samsung 1 TB DDR4
- SSD: Samsung 3.2 TB NVMe

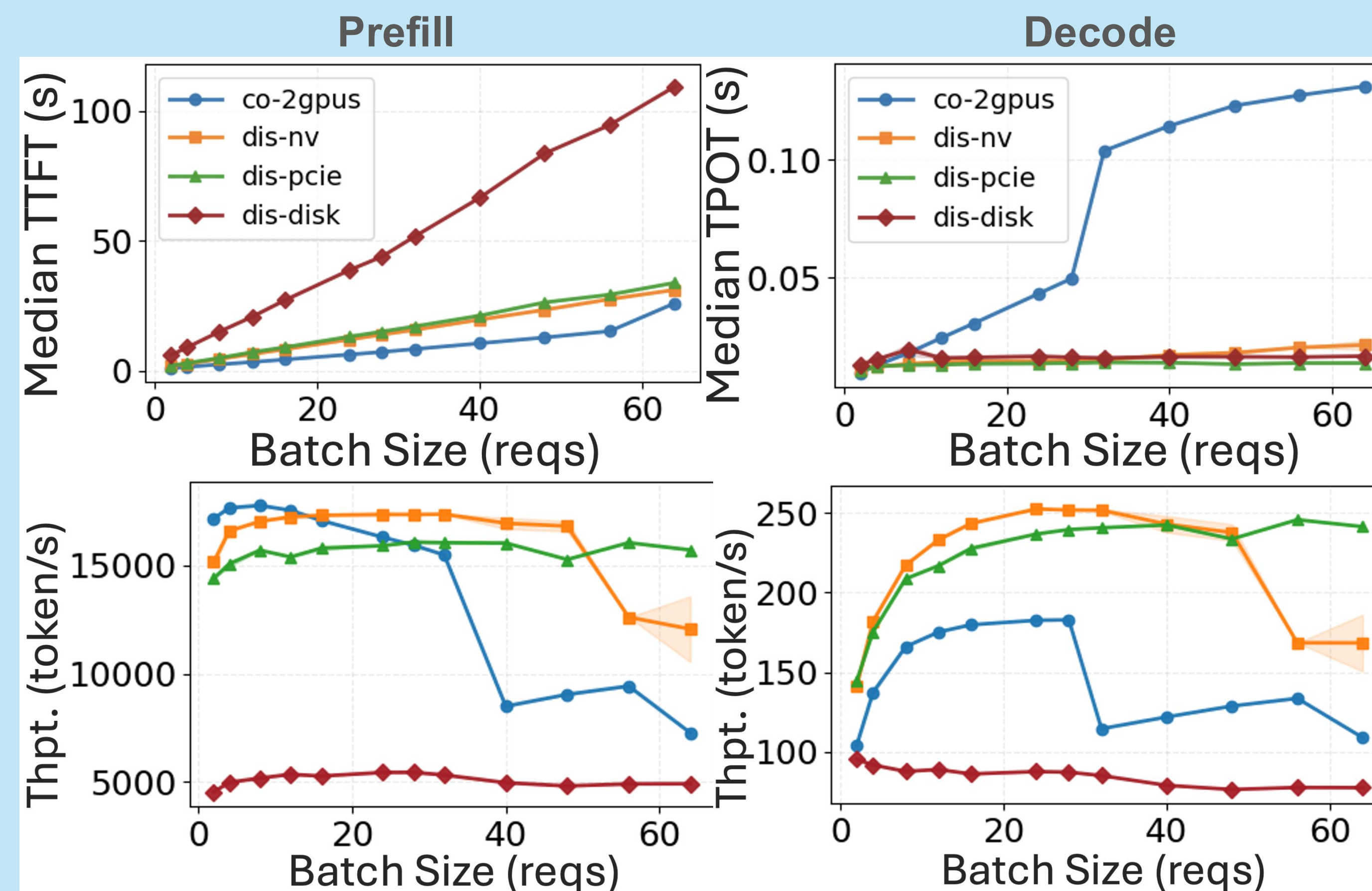
Model and Dataset

- LLM Model: Llama-3.2-3B
- Dataset: Synthetic requests of configurable batch size, input length, and output length (RandomDataset)



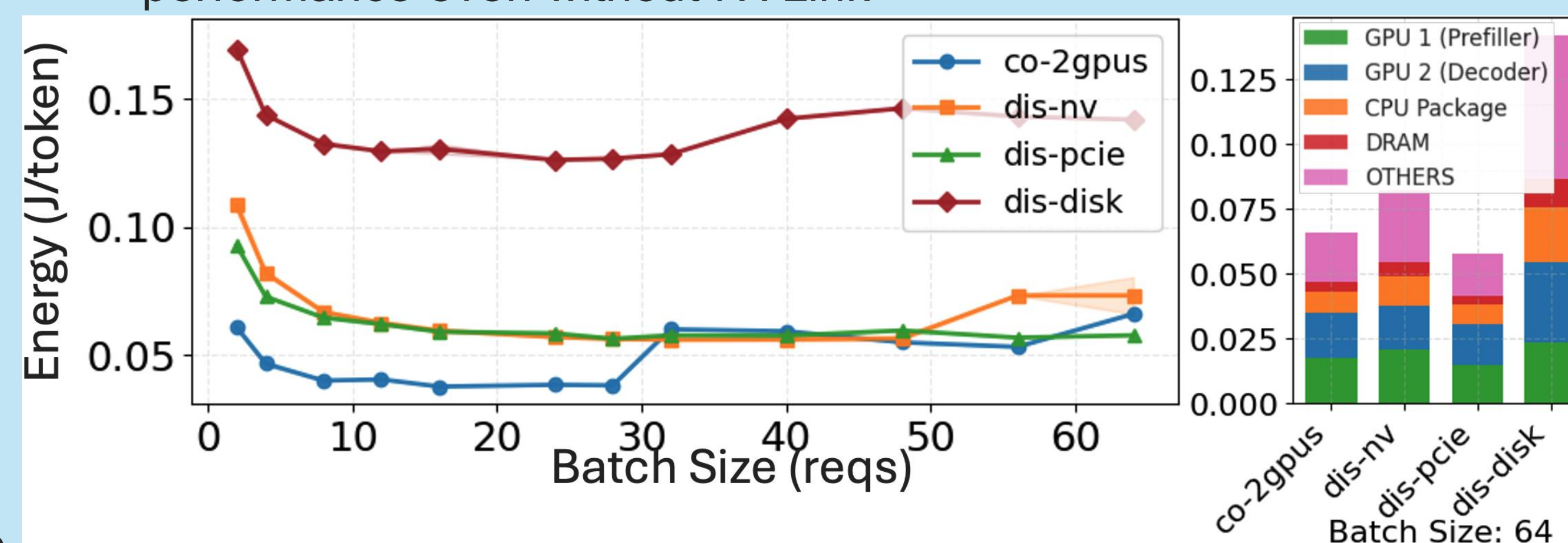
Disaggregated Configurations (2-4)

EXPERIMENT 1: PERFORMANCE AND ENERGY



Observation & Takeaway

- Performance benefits are not universal for prefill–decode disaggregation
 - Effectiveness depends on workload characteristics (e.g., batch size) and inference stage (prefill versus decode)
- KV transmission bandwidth exhibits diminishing returns in performance improvement
 - PCIe-based GPU P2P KV transfer can achieve competitive serving performance even without NVLink



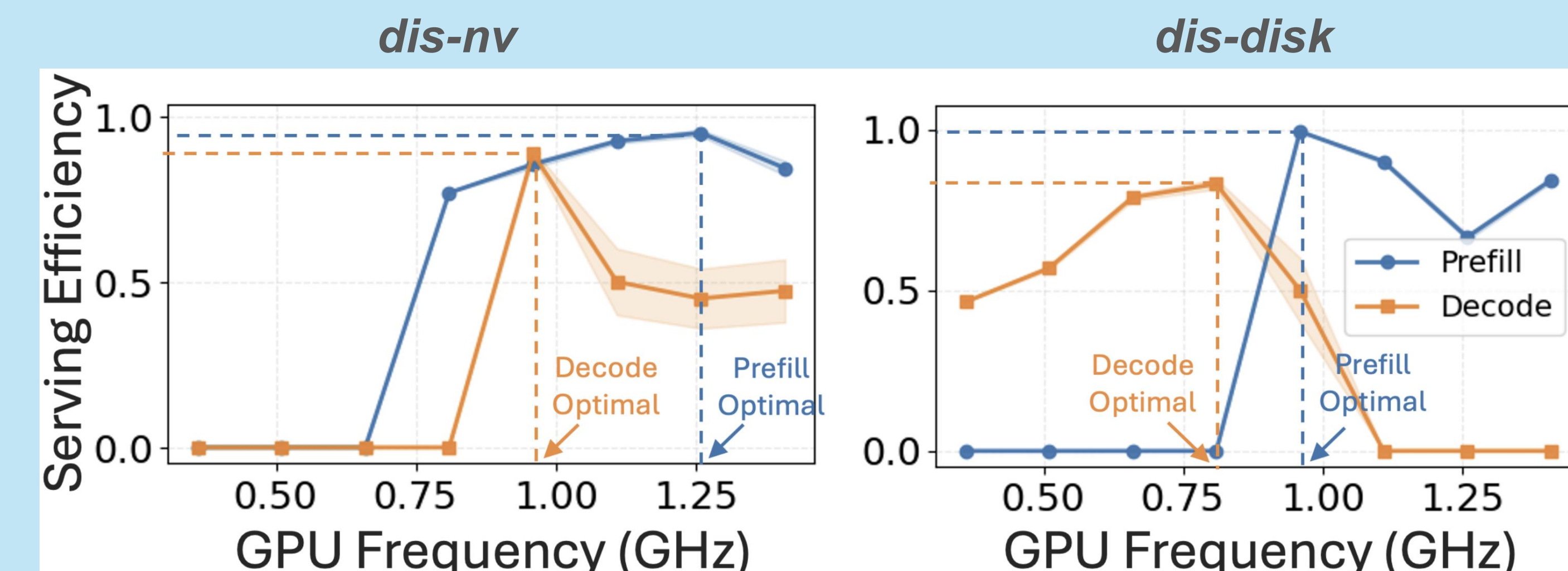
Observation & Takeaway

- Energy consumption from non-GPU components (e.g., CPU, DRAM, and disk) increases as KV transfer path touches deeper memory tiers

REFERENCES

- [1] Chao Wang et al. 2025. Prefill-Decode Aggregation or Disaggregation? Unifying Both for Goodput-Optimized LLM Serving. arXiv:2508.01989.
- [2] Yinmin Zhong et al. 2024. DistServe: Disaggregating Prefill and Decoding for Goodput-optimized Large Language Model Serving. OSDI 24.
- [3] Ruoyu Qin et al. 2025. Mooncake: Trading More Storage for Less Computation — A KVCache-centric Architecture for Serving LLM Chatbot. FAST 25.
- [4] Jiachen Liu et al. 2024. Andes: Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services. arXiv:2404.16283.
- [5] Jovan Stojkovic et al. 2025. DynamoLLM: Designing LLM Inference Clusters for Performance and Energy Efficiency. HPCA 25.

EXPERIMENT 2: DISAGGREGATED GPU FREQUENCY SCALING



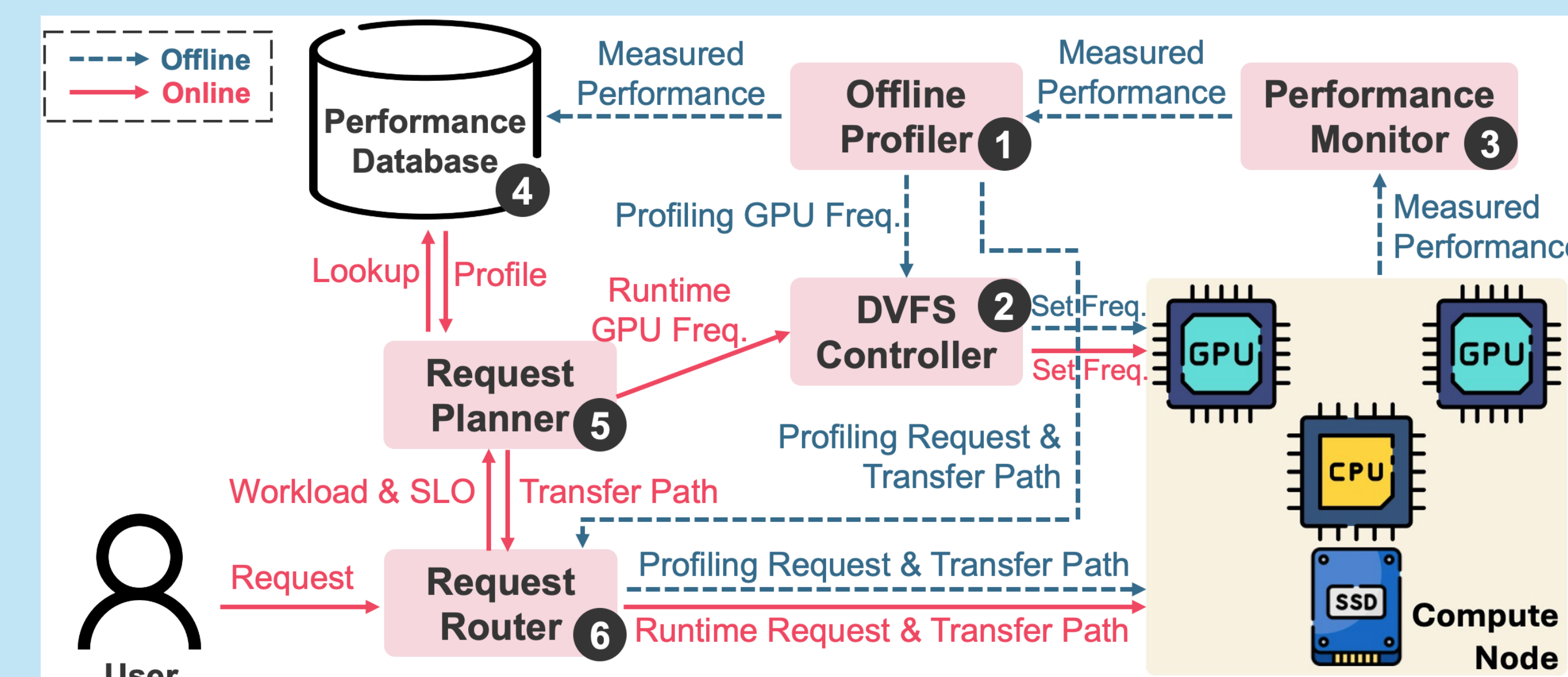
Define:

$$\text{ServingEfficiency} = \begin{cases} \frac{\text{Throughput}}{\text{Energy}}, & \text{Latency} \leq \text{SLO} \\ 0, & \text{Latency} > \text{SLO} \end{cases}$$

Observation & Takeaway

- Optimal GPU frequency can differ between prefill and decode stages
- Independent frequency scaling can further optimize serving efficiency while satisfying the SLO constraints

PHOTOTYPE: SLO-AWARE ENERGY-EFFICIENT LLM SERVING



Offline Profiling Stage

- Offline Profiler 1 performs three-dimensional profiling by varying 1) workload, 2) KV transfer path, and 3) GPU frequency in each iteration
- Performance Monitor 3 records latency, throughput, and energy usage for each iteration and stores them in the Performance Database 4

Online Serving Stage

- Request Router 6 sends the workload characteristics and SLO requirements of queued requests to the Request Planner 5
- By looking up from the Performance Database 4, Request Planner 5 selects the optimal transfer path and frequency configuration that maximizes ServingEfficiency under per-stage SLO constraints

CONCLUSION

- Performance gains of prefill–decode disaggregation are not universal
- Stage-wise frequency scaling enables improved energy efficiency
- Serving configurations should be adaptively determined at runtime